

Script 5: Hyperparameter Summary — Every Choice With Its Research Backing
DP-QLoRA ($\epsilon=1.9911$) · Qwen2.5-1.5B · r=16 · C=1.5 · B=24 · 6 epochs

J: Hyperparameter Justification Table
Every choice + its research backing

Hyperparameter	Value	Justification	Reference
ϵ (privacy budget)	1.9911	Knee of utility curve; Tier 2 strong-end ($e^2 \approx 7.4\times$ vs $e^4 \approx 54\times$ per record)	Ponomareva et al. 2023 §5.2
δ (failure prob)	$2.5 \times 10^{-4} \approx 1/N$	Expected failures < 1 record (4382 train)	Dwork & Roth 2014
C (clip norm)	1.5	Bounds per-record sensitivity; noise $\propto \sigma C$; 1.5 = standard default at this scale	Abadi et al. CCS 2016
B (batch size)	24	Memory limit on T4 16GB with 4-bit+LoRA; NSR=0.213 (acceptable for task)	Ponomareva et al. §5.4.1
Epochs	6	ϵ budget fully spent; loss converged at ep3 (val_loss plateau visible)	DP_logs epochs 1–6
r (LoRA rank)	16	$\alpha/r=2$, q/k/v/o targets; 4.4M trainable (0.49% of 1.5B = minimal DP noise)	Hu et al. ICLR 2022
α (LoRA alpha)	32	$\alpha=2r$ doubles adapter LR — extraction tasks benefit from faster adapter convergence	Biderman et al. TMLR 2024
MAX_LENGTH	64	Compliance requirement; prompt+response fits within 64 tokens	Project constraint
Accounting	RDP via Opacus	Tight per-step composition; auto σ via make_private_with_epsilon()	Mironov CSF 2017
Sampling	Fixed batches (not Poisson)	Required for 4-bit kernel compatibility; ϵ is conservative approximation	Ponomareva et al. §4.3